

PG Management – System Blueprint

1. Product Vision

PG Management is a **subscription-based, multi-tenant SaaS platform** designed to digitize and centralize the management of Paying Guest (PG) accommodations.

Core Objectives

- Enable **PG Owners** to manage properties, rooms, tenants, billing, staff, and analytics.
 - Enable **Tenants** to discover PGs, book rooms, pay rent, raise complaints, and manage their stay digitally.
 - Operate as a **marketplace model**, where **only subscribed PGs are publicly visible**.
-

2. Target Users

Primary Users

- **PG Owners** (Primary revenue source)

Secondary Users

- **PG Tenants** (Free users)
-

3. Revenue Model

- Monthly subscription plans
- Yearly subscription plans (discounted)
- Free trial support (time-bound)

Revenue is generated **only from PG Owners**.

4. User Roles & Access Control (RBAC)

Role Definitions

Role	Access Scope
Super Admin	Platform control, subscriptions, city rollout
PG Owner	Manage PGs, rooms, tenants, billing, staff

Staff	Assigned tasks only
Tenant	PG discovery, booking, payments, complaints

Access Model

- **Single mobile app**
 - **Role-based login and navigation**
 - JWT-based authentication
 - RBAC middleware enforced at API level
-

5. High-Level System Architecture

Client Layer

Mobile Application (Flutter / React Native)

- PG List View
 - PG Map View (OpenStreetMap)
 - Owner Dashboard
 - Tenant Dashboard
-

Backend Services

- Authentication & RBAC Service
 - Subscription Management Service
 - PG & Property Management Service
 - Location & Geo Query Service
 - Tenant & Billing Service
 - Notification Service
-

Data Layer

Database

- PostgreSQL
- PostGIS (for geolocation & map-based search)

Key Data Elements

- org_id (multi-tenant isolation)
 - latitude & longitude
 - geo-indexes for radius search
-

External Integrations

- Payment Gateway: Razorpay (India), Stripe (Global)
 - Maps: OpenStreetMap (via Leaflet / MapLibre)
 - Push Notifications
-

6. Core Functional Modules

1. Authentication & Subscription
 2. PG & Property Management
 3. Room, Bed & Availability Management
 4. Tenant & KYC Management
 5. Billing & Payments
 6. Staff & Complaint Management
 7. PG Discovery (List + Map View)
 8. Reports & Analytics
 9. Notifications & Alerts
-

7. Database Design – Location Enabled

pg_properties Table

Column	Description
pg_id	Primary Key
org_id	Owner Organization
name	PG Name
address	Full Address
city	City
latitude	Geo Latitude
longitude	Geo Longitude

subscription_status	Active / Expired
status	Draft / Published

PostGIS Usage

- Nearby PG search
- Radius-based filtering (e.g., 2 km / 5 km)
- Map-based listing

8. Multi-Tenant Database Architecture

Core Concept

- Each **PG Owner = One Organization (OrgID)**
- Every transactional table includes **org_id**
- Tenant data is strictly isolated

Example Tables

organizations (org_id, owner_id, subscription_plan, status)

pg_properties (pg_id, org_id, name, location)

rooms (room_id, pg_id, category)

beds (bed_id, room_id, status)

tenants (tenant_id, org_id, bed_id)

payments (payment_id, tenant_id, status)

Benefits

- ✓ Secure
- ✓ Scalable
- ✓ Industry standard (Zoho / Odoo SaaS model)

9. PG Owner – Functional Flow

A. Onboarding & Subscription Flow

Owner Signup

- Choose Subscription Plan
- Payment Success
- OrgID Created
- Owner Dashboard Activated

**Business Rule:**

If subscription expires → PG is automatically hidden from marketplace.

B. PG Setup Flow

Create PG

- Add Facilities & Rules
 - Add Floors → Rooms → Beds
 - Upload Room Images
 - Set Rent & Deposit
 - Publish PG
-

C. Tenant Lifecycle Flow

Tenant Applies

- Owner Approval
 - Room / Bed Allocation
 - Check-in
 - Monthly Rent Cycle
 - Notice Period
 - Check-out & Deposit Refund
-

D. Billing & Finance Flow

Monthly Invoice Auto-Generated

- Tenant Pays (UPI / Card / Wallet)
- Receipt Generated
- Owner Dashboard Updated

Supports:

- Electricity charges
 - Food charges
 - Wi-Fi charges
 - Penalties
-

E. Staff & Complaint Flow

Tenant Raises Complaint

- Assigned to Staff

→ Status Update (Open → In Progress → Closed)

10. Tenant (Customer) Experience Flow

PG Discovery (Marketplace)

Filters available:

- Location
- Rent
- Sharing type
- Food preference
- AC / Non-AC

 Only PGs with active subscriptions are shown

Booking Flow

Select PG

→ View Rooms & Photos

→ Apply

→ Owner Approval

→ Pay Advance

Stay Management

- Monthly rent payment
- Download receipts
- Raise complaints
- Chat with PG owner
- Checkout request

11. Subscription Model (PG Owner)

Plan	PG Limit	Tenant Limit
Basic	1	20

Standard	5	100
Premium	Unlimited	Unlimited

Features:

- ✓ Razorpay / Stripe
- ✓ Auto-renew
- ✓ Trial support

12. Technology Stack (Final Recommendation)

Frontend

- Android, iOS, future Web
- Push Notifications

Backend

- Node.js (NestJS) or Django
- JWT Authentication
- RBAC Middleware

Database

- PostgreSQL
- PostGIS
- Row-level tenant isolation

Infrastructure

- AWS EC2 / RDS
- S3 for images & documents
- CloudWatch for logs & monitoring

13. Roadmap

Play Store Launch

- PG listing (List + Map)
- Room & bed management
- Tenant management
- Rent tracking

ZYVONE TECHNOLOGY

- Subscription system

Phase-2 Enhancements

- Staff attendance
- Tenant KYC verification
- Advanced analytics dashboard
- Ratings & reviews
- Visitor management